

Question 12

```
vedsk0831@DESKTOP-83FU7HB:~$ cd ~/ossp
nano q12.c
vedsk0831@DESKTOP-83FU7HB:~/ossp$ gcc -Wall -Wextra q12.c -o q12
vedsk0831@DESKTOP-83FU7HB:~/ossp$ ./q12
Pipe created successfully.
Parent: Data written to pipe.
Child: Reading data from pipe...
Child received: Hello from parent process!
Parent: Child process completed.
vedsk0831@DESKTOP-83FU7HB:~/ossp$ strace ./q12
execve("./q12", ["/q12"], 0x7ffd9c9c67e0 /* 27 vars */) = 0
brk(NULL) = 0x55cb196fd000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x76699d9a3000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20811, ...}) = 0
mmap(NULL, 20811, PROT_READ, MAP_PRIVATE, 3, 0) = 0x76699d99d000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0"... , 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2190608, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0"... , 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76699d600000
mmap(0x76699d628000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|M
```

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

 $\{$

```
int pipefd[2];
```

```
/* Create pipe */
if (pipe(pipefd) == -1)
{
    perror("pipe");
    return 1;
}

printf("Pipe created successfully.\n");

/* Create child process */
pid_t pid = fork();

if (pid < 0)
{
    perror("fork");
    return 1;
}

/* Child process */
if (pid == 0)
{
    close(pipefd[1]);

    /* Redirect pipe input to stdin */
    dup2(pipefd[0], STDIN_FILENO);

    close(pipefd[0]);
```

```
printf("Child: Reading data from pipe...\n");
```

```
char buffer[100];
```

```
ssize_t n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);
```

```
if (n > 0)
```

```
{
```

```
    buffer[n] = '\0';
```

```
    printf("Child received: %s\n", buffer);
```

```
}
```

```
exit(0);
```

```
}
```

```
/* Parent process */
```

```
else
```

```
{
```

```
    close(pipefd[0]);
```

```
char message[] = "Hello from parent process!";
```

```
/* Write data into pipe */
```

```
write(pipefd[1], message, sizeof(message));
```

```
printf("Parent: Data written to pipe.\n");
```

```
/* Close write end */  
close(pipefd[1]);  
  
/* Wait for child */  
waitpid(pid, NULL, 0);  
  
printf("Parent: Child process completed.\n");  
}  
  
return 0;  
}
```